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PROJECT REPORT ON HOTEL NETWORK

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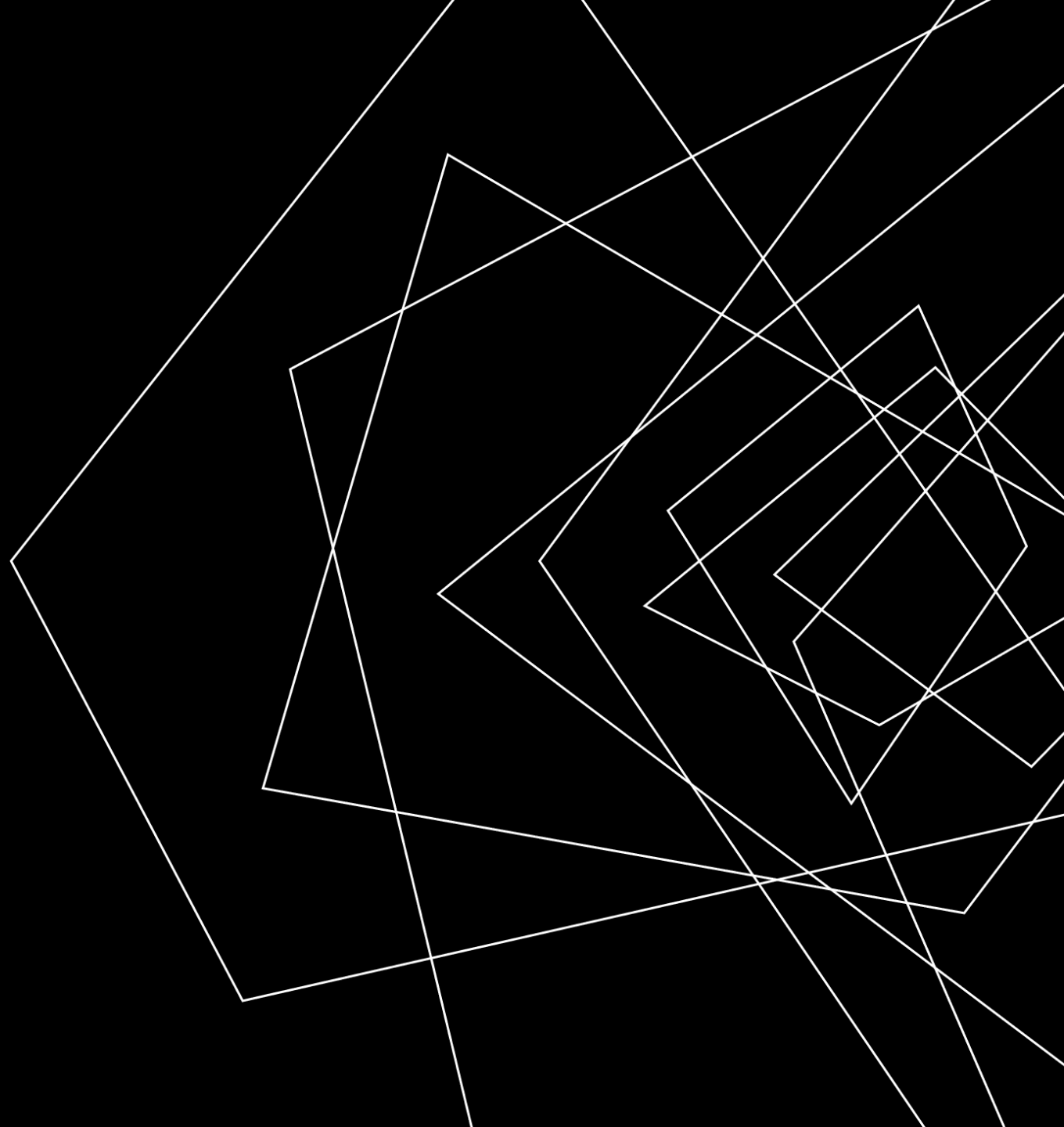


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At last, I would like to extend my heartfelt thanks to my parents because without their help this project would not have been successful. Finally, I would like to thank my dear friends who have been with me all the time.

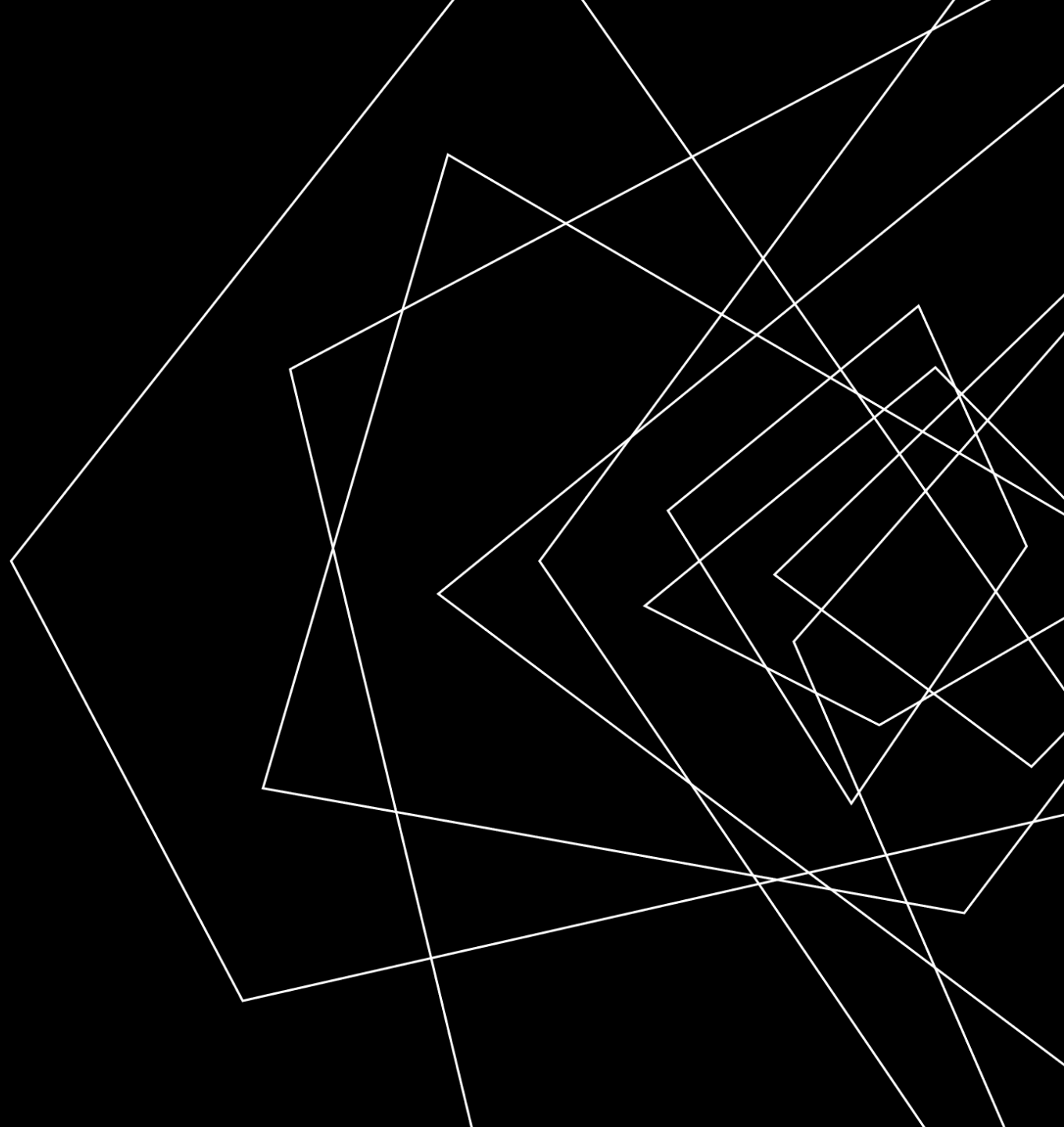


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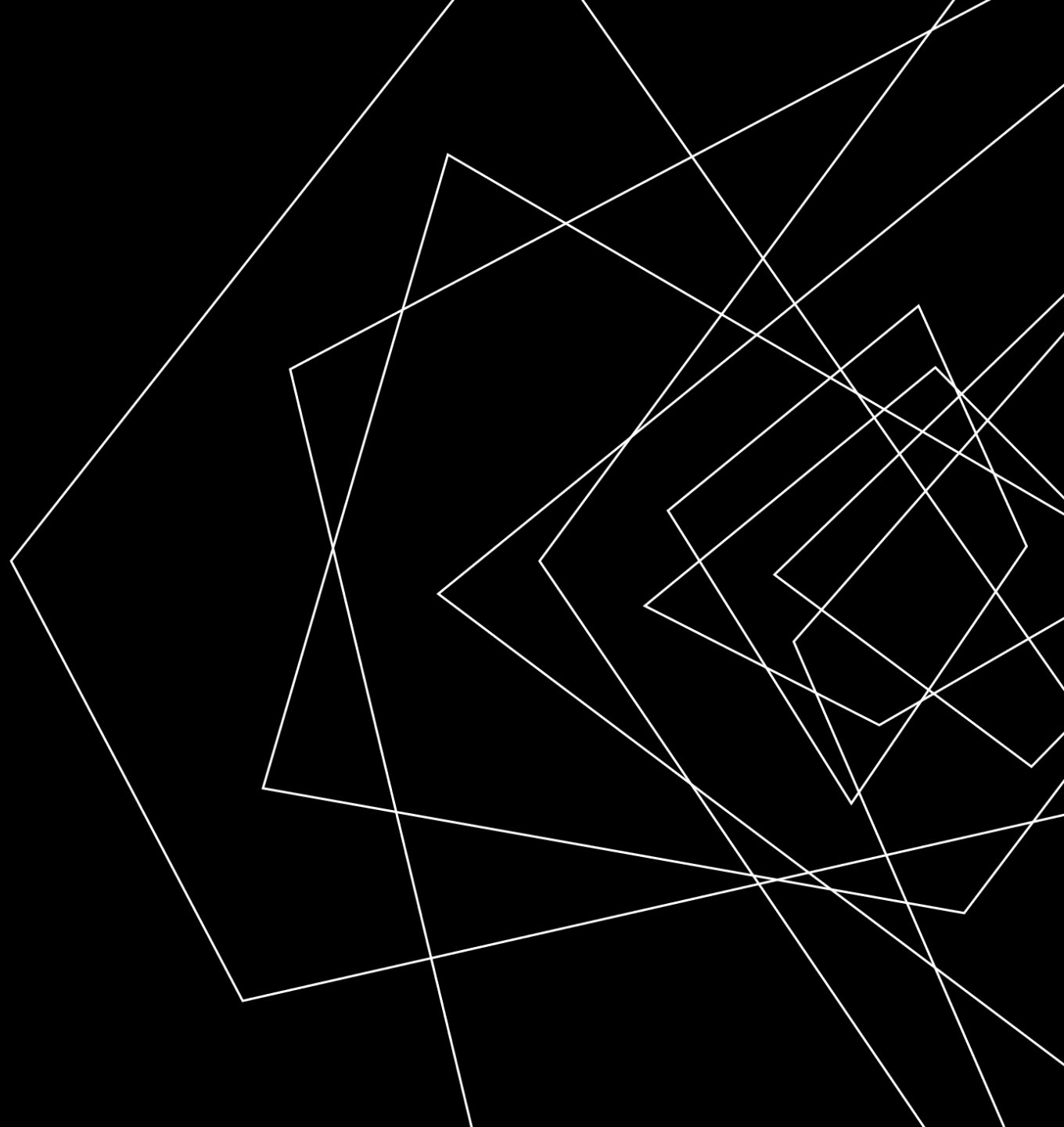
INTRODUCTION ON HOTEL NETWORK

Computer network borne out of the need to use personal computers for sharing information within an organization in the form of messages, sharing files, resources and so forth. Whether the organization is located in a small area like as in one building or spread over a large area, the need for networking the computers cannot be over emphasized. A Hotel area network is an essential part of Hotel system. A Hotel network has several uses such as, result publishing, resource sharing, file sharing, communication, etc. The Hotel Area Network design is about designing a topology of a network that is a LAN (Local Area Network) for a Hotel in which various computers of different departments and buildings are Setup so that they can interact and communicate with each other by interchanging data. To design a network for a Hotel, which connects various departments and buildings to each other



HOTEL NETWORK REQUIREMENTS

- Provide powerful wired and wireless networks to both customers and employees.
- Complete hotel WiFi coverage in all areas, including outdoor areas such as parking and swimming pools.
- High network stability and security to ensure uninterrupted business and guest data security.
- Offer advertising and promotional content to affiliate customers.
- Elegant appearance to blend with hotel decoration.
- Simple hotel network management and convenient equipment maintenance



HOTEL WIFI NETWORK DESIGN BASICS

From site assessments to proper configurations, each element of your hotel WiFi network must fit into your budget and hotel design while meeting current and future guest expectations. Your provider will determine the best access points (AP) for your hotel, give you equipment options, and suggest ways to achieve your network goals. So, [don't make these five hotel WiFi design mistakes](#).

Equipment quality and placement also play a significant role in your hotel network. Providers help you choose high-quality routers and access points. While most AP's use layer 2 isolation, you'll see many differences in equipment like:

- Integrated, stock, directional, or external antennas
- Rated for indoor or outdoor use
- Total antenna output measured in decibels relative to isotropic (dBi)
- Service set identifier (SSID) and wireless distribution system (WDS) capabilities

HOTEL WIFI NETWORK DESIGN: INSTALLATION STRATEGIES

There is no one-size-fits-all managed network WiFi solution for hotels. Your installation strategy depends on ideal AP placement, coverage, capacity, and channel configuration. Work with a managed WiFi network provider to assess your existing infrastructure and determine your needs. Doing so helps you get the best outcome and provides options for scaling up or down.

Site Survey and Pre-Design Assessments

During a site survey, your provider completes a thorough inspection of your property and current network infrastructure. They note the property construction, such as the use of brick, concrete, or drywall. Each element affects signal penetration. A site survey also identifies:

- Potential access point locations
- Line of sight viability
- All essential coverage areas

Network engineers use site survey data to develop an equipment list and installation diagram. They look at overall design by considering how your AP signals overlap, determine the ideal WiFi network configuration, and choose appropriate channels. Proper AP placement delivers reliable coverage with minimal interference.

Consultation for Requirements and Design Considerations

Each aspect of your network infrastructure must meet your current needs. But, it's also vital to future-proof your hotel by planning for growth and [the future of technology](#). Along with a site survey, your provider assesses requirements and design considerations for:

- Service and access
- Roaming
- Security
- Management
- Aesthetics
- Operations and maintenance (O&M) program
- Authentication
- Reliability

Your managed network provider uses data to help you select the best installation plan and equipment. A hotel WiFi network diagram maps out your WiFi routers, access points, controllers (also called gateways), and switches.

You may need extra cabling to support your network, as a wireless distribution system may not perform well for all hotel guests. In many cases, your network diagram will include a wireless range extender, also called a wireless repeater, to bridge infrastructure gaps and rebroadcast the signal from your AP or router.

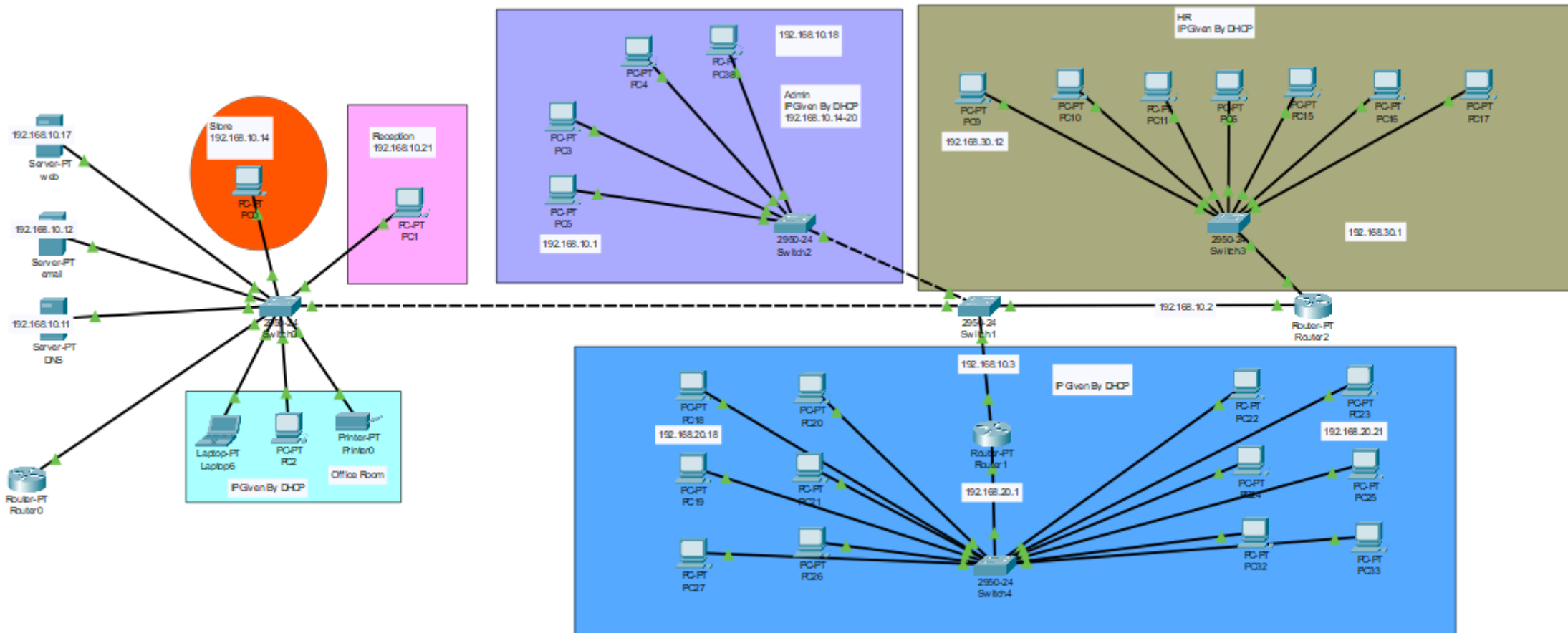
Hotel WiFi Installation and Integrations

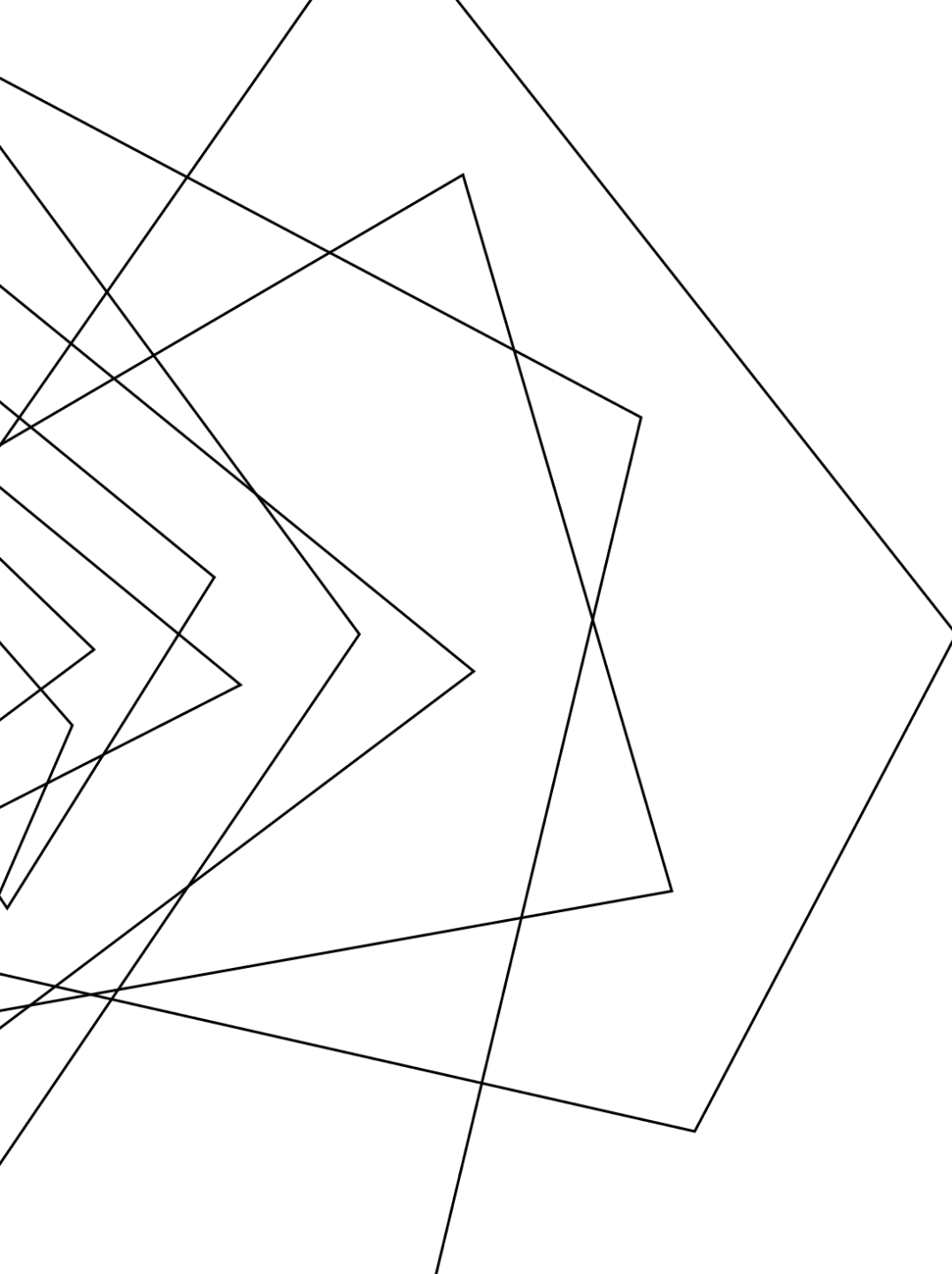
When possible, complete your hotel WiFi set up during downtime. Doing so avoids disturbing guests and allows for staff training. Using a WiFi network diagram, installers place, connect, and test infrastructure and equipment. If you have third-party signage systems or other software, your managed network provider installs this after your network is running.

After Installation Support and Analysis

Get the most from your managed network solution by surveying after installation. Your provider analyzes data from each access point to ensure it meets data minimums while identifying interference and channel availability. This helps you optimize your configuration and learn how the system functions.

BASIC ARCHITECTURE





CONCLUSION

To Design the network outlook for the community Hotel network scenario produces the substructure for all other exposure in the service framework such as security of the network, wireless area network, mobility as well as putting the justification to provide safety and security, operational efficiencies, virtual environments, and secure rooms. This paper describes the network design scenario approved by Cisco, as well as where we can apply these scenario within the various locations of a community Hotel network. Finally, key network foundation services such as switching, routing, multicast, and high availability are given for the full Hotel network scenario.